

Detecting and mitigating IMSI catcher attacks: A formalized framework for secure mobile networks

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ABSTRACT

Despite their universal integration into our daily life, mobile communication networks remain susceptible to sophisticated security threats. Among these threats, the IMSI (International Mobile Subscriber Identity) catcher, which stands out as a stealthy device capable of intercepting mobile communications, tracking user locations, and compromising the users privacy. Traditional solutions against this threat often rely on detecting anomalies in individual base station parameters, which can be easily mimicked by advanced IMSI catchers. This paper provides a novel IMSI catcher detection method that leverages the unique characteristics of cellular location areas, rather than individual base stations, to create a robust "cellprint". Building upon the foundational concept of location area features, this work provides an enhanced theoretical analysis of cellprint generation, including detailed mathematical modeling and formal proofs of its robustness against various attack vectors. Furthermore, we present a simulation-based validation of the proposed method, evaluating its detection rate, accuracy, latency, and overhead under diverse network conditions and against different IMSI catcher behaviors. Our findings demonstrate that the location area fingerprinting approach offers a resilient and scalable solution for detecting IMSI catchers, significantly raising the bar for adversaries. This research contributes to the ongoing efforts to strengthen cellular network security in an era of evolving threats and increasingly complex network architectures, including considerations for 5G and 6G.

1. Introduction

The ubiquitous nature of mobile communication has transformed global connectivity, yet it simultaneously introduced a countless number of security vulnerabilities that threaten user privacy and data integrity (Kareem, 2024; Arapinis et al., 2017; Steig et al., 2016). The International Mobile Subscriber Identity (IMSI), is a unique identifier for each Subscriber Identity Module (SIM) card. IMSI contains a 15-digit number used to identify the mobile user on the cellular network in any place in the globe (Steig et al., 2016). This number is divided into three parts:

- Mobile Country Code (MCC), 3 digits.
- Mobile Network Code (MNC), 2/3 digits.
- Mobile Subscriber Identity (MSI), 9/10 digits.

It is a critical component in cellular networks, which enable the identification and authentication of mobile devices. However, the exposure of the IMSI during initial network attachment and location updates

has been a significant security concern, exploited primarily by devices known as IMSI catchers (Ludant et al., 2024). IMSI catcher, often referred to as a StingRay or a rogue base station, it is a radio device designed to mimic a legitimate cellular base station. By broadcasting a stronger signal or exploiting vulnerabilities in cellular protocols, it tricks nearby mobile phones into connecting to it instead of the authentic network infrastructure (Ludant et al., 2024). Once a mobile device connects to an IMSI catcher, the attacker gains the ability to intercept calls and SMS messages, track location, and potentially inject malicious content or downgrade network services (Hussain et al., 2019). While they were initially developed and used by government agencies for law enforcement and national security purposes, the increasing availability of these devices on the commercial market poses a growing threat to civil liberties and personal privacy (Hussain et al., 2019). Several countermeasures against IMSI catchers have been proposed in the literature, but most existing solutions rely on mobile tower characteristics to differentiate legitimate base stations from fake ones. However, these

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characteristics are limited and can be easily mimicked by IMSI catcher devices.

This research introduces a novel approach that uses comprehensive location area features to detect and mitigate the presence of IMSI catchers in a specific location area. It builds upon the foundational work presented in Alrashde and Shaikh (2019), which introduced a novel concept of detecting IMSI catchers by analyzing the aggregate features of an entire cellular location area, rather than relying solely on individual base station parameters. The core idea, termed “cellprint”, assumes that the unique combination of base stations within a geographical location area creates a distinct fingerprint that is difficult for a single rogue device to replicate without altering the overall characteristics of that area. The original work proposed two methods for cellprint generation: CID-based and location-based, and qualitatively discussed their potential. The contributions of this research include providing a more rigorous theoretical foundation and a complete simulation-based validation. Specifically, this paper focuses on:

1. **Enhanced Theoretical Analysis and Formalization:** We delved deeper into the mathematical modeling of cellprint generation, exploring advanced schemes for CID-based cellprints and formalizing the topological properties for location-based cellprints. We also provided formal arguments for the robustness of the cellprint concept against sophisticated IMSI catcher evasion techniques.
2. **Simulation-Based Validation:** We established a realistic simulation environment to quantitatively evaluate the proposed method. This includes assessing its detection rate, false positive/negative rates, detection latency, and the computational and communication overhead under various network conditions and IMSI catcher behaviors. This empirical validation provides crucial insights into the practical applicability and performance of the cellprint approach.

The rest of the paper is structured as follows. Section 2 reviews related work, countermeasures, and identifies the research gap which addressed in this study. Section 3 introduces the IMSI catcher threat model and its attack scenarios. Section 4 presents the methodology and the proposed CellPrint framework. Section 5 details the experimental setup, while Section 6 explores the results and performance evaluation. Section 7 discusses the implications of the findings, acknowledges the study’s limitations, and explores the future research directions. Finally, Section 8 concludes the paper.

2. Related work and countermeasures

Lots of work and countermeasures have been conducted by researchers and mobile companies to detect and mitigate the threats of IMSI catcher. This section explores these efforts and compares between them. Initial research focused on demonstrating the feasibility of IMSI catcher attacks and developing foundational detection techniques. Suraev (2014) demonstrated how low-cost software-defined radios (SDRs) could be used to build functional IMSI catchers, underscoring the urgent need for defensive strategies. Later on, Chlosta et al. (2021) demonstrated a “SUCI-Catcher” attack, proving that even the 5G Subscription Concealed Identifier (SUCI) is vulnerable to linkability attacks that can deanonymize users. Preuß Mattsson and Nakarmi (2021) further analyzed the SUCI mechanism, concluding that it can provide poor anonymity and recommending padding schemes to strengthen it. In response, researchers began developing detection and prevention mechanisms.

Dabrowski et al. (2014) proposed a solution using mobile applications and stationary sensors to identify malicious network behavior. This approach was later extended toward operator-based monitoring, which leveraged network-side data to detect threats beyond the user’s vicinity (Dabrowski et al., 2016). These pioneering works illustrated the

dual potential of both end-user tools and infrastructure-level solutions but also faced challenges, such as limited detection accuracy or the need for deep integration with network operators. Further highlighting the need for robust detection, Park et al. (2019) conducted a systematic study of commercial IMSI catchers, analyzing their attack techniques and the vulnerabilities they exploit across 2G, 3G, and 4G networks.

Subsequent efforts turned to machine learning (ML) to enhance the accuracy and robustness of detection. Al Odaini et al. (2023) demonstrated that convolutional and recurrent neural networks could effectively identify the distinct behavioral patterns of IMSI catchers with high precision. Mubasshir et al. (2024) developed FBSDetector, an Android-based tool that achieved over 90% accuracy in real-world tests by using ML to detect fake base stations and more complex multi-step attacks. Similarly, Nakarmi et al. (2022) applied ML models to features derived from the Reference Signal Received Power (RSRP), demonstrating a method for detecting false base stations without needing prior data from malicious sources for training.

Other innovative approaches combined ML with additional data sources. Richter et al. (2024) introduced a multi-modal detection system that integrated geolocation data and hardware fingerprinting, showing that physical-layer information can substantially improve detection robustness. Further expanding on this, Kim et al. (2024) proposed a detection and blacklisting scheme that identifies fake base stations by monitoring real-time network information without altering existing cellular protocols. At a larger scale, the SeaGlass project described by Ney et al. (2017) deployed networks of low-maintenance sensors to identify rogue base stations across a city, though such approaches involve significant deployment costs.

Another critical line of research has addressed IMSI catcher threats from the protocol side by enhancing cryptographic protections. Khan et al. (2018) proposed a scheme combining pseudonyms with public-key encryption to mitigate downgrade attacks, while Norman et al. (2016) introduced pseudonym-based methods compatible with existing SIM cards, although these approaches suffered from synchronization issues. Zhang et al. (2023) suggested improvements to the 5G Authentication and Key Agreement (AKA) protocol by incorporating asymmetric encryption and stronger authentication features. By looking toward future threats, Ulitzsch et al. (2022) proposed post-quantum secure SUCI schemes to provide long-term resilience. Other studies focused on end-to-end encryption, such as the CL-PKC-based scheme for GSM developed by Ramadan et al. (2016) and the 5G protection framework by Ali et al. (2022). While the mentioned solutions are powerful, many of these cryptographic enhancements would require significant modifications to both network infrastructure and user devices.

Finally, research into physical-layer attacks has highlighted the increasing sophistication of modern IMSI catchers. Kotuliak et al. (2022) presented LTrack, a passive LTE-based tracking method capable of extracting a user’s IMSI and location with high accuracy without active transmissions, making it difficult to detect. Erni et al. (2022) introduced AdaptOver, an “overshadowing” man-in-the-middle attack that exploits signaling procedures in both LTE and 5G. These studies emphasize the need for detection methods that can withstand adaptive and stealthy adversaries. Table 1 provides a comparative overview of the discussed articles, detailing their primary methods, advantages, and disadvantages.

Countermeasures against IMSI catcher threats include open-source software mobile applications that can analyze phone connections and detect IMSI catcher devices. These applications differ in some features and share similarities in others (Park et al., 2017; Android IMSI-Catcher Detector, 2025). Table 2 shows a brief about those applications.

Mobile phone companies have also introduced secure and encrypted devices as a response to the threat of IMSI catcher. Some rely on landlines, some on satellites, and others use VPNs to protect communication (Cryptophone Mobile, 2025; XCell Stealth Phones, 2025; SGP Black-Phone, 2025; Lance Whitney, 2025). Table 3 shows a comparison between the various secure mobile phones.

Table 1

A comparative overview of the related studies, outlining their approaches, advantages, and limitations.

Reference	Method	Advantages	Disadvantages
Chlosta et al. (2021)	Proof-of-concept “SUCI-Catcher” attack in a 5G standalone network.	Demonstrates that 5G’s encrypted identifiers can still be linked, highlighting a key vulnerability.	The proposed countermeasures require implementation by network operators.
Preuß Mattsson and Nakarmi (2021)	Analysis of 5G SUCI mechanism vulnerabilities.	Identifies weaknesses in the 5G standard’s privacy features and proposes practical improvements (padding).	The proposed fixes require adoption by network operators.
Dabrowski et al. (2016)	Mobile applications, stationary sensors, and operator-based monitoring for detection.	Explored both end-user and infrastructure-level solutions for broad detection coverage.	Limited accuracy; operator-based solutions require network integration.
Park et al. (2019)	Systematic study of commercial IMSI catchers and analysis of their attack techniques.	Offers a detailed analysis of real-world attacker capabilities and highlights limitations of current detectors.	Focuses on analysis rather than proposing a new detection mechanism.
Mubasshir et al. (2024)	Machine learning models (FBSDetector) on the user device to detect fake base stations.	Achieves high detection accuracy with low false positives using a dedicated dataset.	Relies on machine learning models that require training and may have performance overhead on the device.
Nakarmi et al. (2022)	Machine learning models using RSRP-based features for false base station detection.	Can detect false base stations without requiring malicious data for training the models.	Real-world performance is a challenge.
Kim et al. (2024)	Detection and blacklisting scheme using real-time network information.	Does not require changes to existing cellular protocols and can defend against Sybil attacks.	Depends on specific network conditions and attacker behavior.
Ney et al. (2017)	City-wide sensor network (SeaGlass) to detect rogue base stations at scale.	Provides wide-area coverage for identifying malicious network infrastructure.	Involves high deployment costs and significant infrastructural overhead.
Khan et al. (2018)	Cryptographic scheme using pseudonyms and public-key encryption.	Aims to mitigate downgrade attacks by strengthening authentication protocols.	Requires modifications to existing network protocols and device capabilities.
Ziayi et al. (2021)	Android-based sensor (YAICD) that analyzes signal parameters to detect MITM attacks.	Can be installed on mobile devices and detects both connected and nearby IMSI catchers.	Limited to GSM networks and requires sensor installation on the device.
Peeters et al. (2023)	Distance bounding over the audio channel (LeopardSeal) to detect call interception.	Novel technique for detecting active call eavesdropping by identifying extra wireless hops.	Specifically designed for detecting audio rogue base stations and may not apply to other IMSI catcher types.
Fraunholz et al. (2022)	Fingerprinting attack exploiting unprotected information in the 4G/5G attach request.	Exposes a fundamental vulnerability in the mobile communication initialization procedure.	Focuses on demonstrating an attack rather than providing a detection solution.
Palamà et al. (2020)	Analysis of handset responses to IMSI catcher attacks across different brands.	Highlights that device behavior is inconsistent, complicating universal detection solutions.	Does not propose a solution but identifies a key challenge for other researchers.

Table 2

A comparison between mobile security apps.

Name	Platform	Functionality
SnoopSnitch	Android	Collects and analyzes frequency and phone data to alert user of malicious activity.
CatcherCatcher	Android	Detects IMSI catcher devices by analyzing broadcast parameters.
IMSI Catcher Detector	Android	Detects IMSI catchers by checking CIDs, LACs, and other features.
Darshake	Android	Monitors silent SMS; shows network state; detects IMSI catchers.
Zimperium zIPS	iOS	Intrusion Prevention System; prevents MitM attacks including IMSI catcher.

Table 3

A comparison between secure mobile phones.

Name	Company	Functionality
CryptoPhone (Crypto-phone Mobile, 2025)	GSMK	End-to-end encryption for all communications.
Stealth Phones (XCell Stealth Phones, 2025)	XCell	Warns users of interceptors (including IMSI catchers); supports dynamic IMEI numbers.
BlackPhone (SGP Black-Phone, 2025)	SGP	Uses VPN for all user communications.
FreedomPop (Lance Whitney, 2025)	Samsung	Encrypts voice/SMS with 128-bit encryption; allows phone number changes.

2.1. Research gaps and motivation for cellprint

Despite significant advancements, several research gaps still persist. Most existing methods focus on anomalies in individual base station parameters. Advanced IMSI catchers can mimic these values, making detection harder. In contrast, replicating the collective characteristics of a geographical area with multiple base stations is much more challenging. Moreover, secure mobile phones are costly and not affordable for most users. This observation forms the core motivation for the location area fingerprinting concept. By focusing on the aggregate features of a cellular location area, the proposed “cellprint” aims to create a robust and difficult-to-spoof identifier. The hypothesis is that an IMSI catcher, when introduced into a location area, will inevitably alter the collective properties of that area in a detectable way, even if it successfully mimics individual base station parameters. Our work builds on this promising concept by establishing the theoretical groundwork and empirical evidence needed for its deployment in 5G and 6G infrastructures.

3. Threat model

The fundamental principle behind an IMSI catcher’s operation is to act as a Man-in-the-Middle (MITM) device, positioning itself between a mobile user’s device (User Equipment—UE) and the legitimate cellular network, as shown in Fig. 1. The primary goal is to force the UE to connect to the rogue device (Park et al., 2019), thereby gaining control over its communication, typically through several mechanisms explained in the next subsections.

3.1. Adversary capabilities

Adversaries range from basic to highly sophisticated operators, such as law enforcement, intelligence agencies, or malicious actors. Their capabilities include:

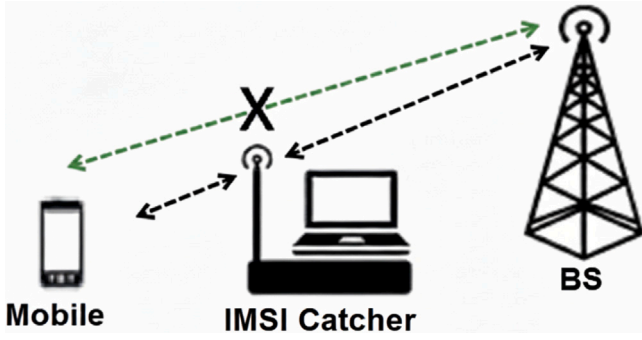


Fig. 1. IMSI catcher threat model.

- **Stronger Signal Emission:** IMSI catchers often broadcast a stronger signal than legitimate base stations in their vicinity, luring mobile devices to connect.
- **Disruption and Evasion:** Includes jamming frequencies to force disconnection from legitimate base stations, protocol downgrades (e.g., LTE to GSM), denial-of-service (DoS), and parameter changes to avoid detection.
- **Impersonation:** Mimicking legitimate base stations (BS) and cloning identifiers such as Cell ID, MNC, and MCC. The IMSI catcher simultaneously acts as a fake UE to the real BS and as a fake BS to the real UE.
- **Data Collection and Interception:** Harvesting IMSIs or SUCIs, intercepting communications via MitM, and extracting metadata like location and device capabilities (Paci et al., 2024).
- **Coordinated Multi-device IMSI Catchers:** Multiple adversaries distributed across a location area trying to compromise the network.
- **Replacement-based Attacks:** Physically replacing legitimate base station hardware or compromising BS operation through network intrusion.

3.2. System assumptions

IMSI catchers exploit inherent vulnerabilities in cellular protocols, particularly in GSM, UMTS, and LTE. A critical weakness in GSM and earlier LTE standards is the unidirectional authentication process: the mobile device authenticates itself to the network, but the network does not authenticate itself to the device. This asymmetry allows an IMSI catcher to impersonate a legitimate BS without being challenged.

3.3. Attack scenarios

Once a UE connects to an IMSI catcher, the attacker can perform various malicious activities:

- **Downgrade Attacks:** Forcing a 4G/3G device to connect via less secure 2G, where IMSI is transmitted in clear text.
- **IMSI/IMEI Harvesting:** Extracting unique identifiers for tracking and surveillance.
- **Call and SMS Interception:** Acting as a relay to record or manipulate communications.
- **Location Tracking:** Continuous monitoring of IMSIs enables fine-grained tracking.
- **Denial of Service:** Selective blocking or degrading of services for users or regions (Ginzboorg and Niemi, 2016; Harvanek et al., 2024).

4. Proposed solution: Cellprint framework

This section provides the theoretical foundation and design of the proposed cellprint framework. It includes mathematical formalization, algorithm for detection, and robustness analysis.

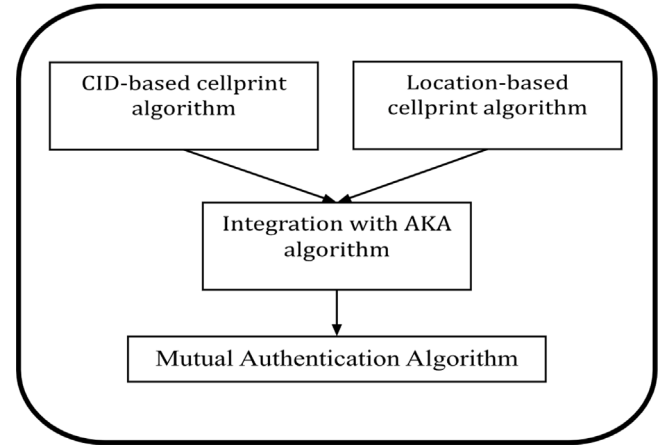


Fig. 2. High-level architecture of the proposed model.

4.1. Framework overview

The proposed model integrates fingerprinting algorithms with the Authentication and Key Agreement (AKA) procedure Fig. 2. This ensures mutual authentication between devices and base stations before connections are established and allows exposure of intruder IDs.

4.2. Mathematical model and formalization

The cellprint, as a unique identifier for a cellular location area, is designed to capture the collective characteristics of the base stations within that area. We formalize two primary methods for cellprint generation: CID-based and location-based, and explore advanced considerations for each. Our method is primarily designed for operator-side deployment (integrated with AKA), rather than purely device-side detection, this significantly mitigates the cold-start problem. The network operator maintains a comprehensive database of cellprints for all location areas under their coverage. This database is continuously updated during normal operation, and user devices query the operator's cellprint database during authentication, rather than independently discovering cellprints.

4.2.1. CID-based cellprint generation

Let LA_k denote a location area with LAC k , containing N_k legitimate base stations BS_i , each with a unique Cell ID CID_i . The CID-based cellprint is calculated using Eq. (1).

$$CP_{CID,k} = \sum_{i=1}^{N_k} CID_i \quad (1)$$

If an intruder with spoofed $CID_{IC,j}$ is added, the altered fingerprint becomes as in Eq. (2).

$$CP'_{CID,k} = CP_{CID,k} + CID_{IC,j} \quad (2)$$

Thus, the intruder ID is exposed using Eq. (3).

$$CID_{IC,j} = CP'_{CID,k} - CP_{CID,k} \quad (3)$$

To improve robustness, a weighted version can be defined as shown in Eq. (4).

$$CP_{CID,k}^{weighted} = \sum_{i=1}^{N_k} w_i \cdot CID_i \quad (4)$$

where w_i could represent signal strength SS_i or historical uptime U_i .

4.2.2. Location-based cellprint generation

The location-based cellprint leverages the geographical distribution and inter-connectivity of base stations within a location area. This method is particularly resilient because altering the physical or logical topology of a network is far more challenging for an IMSI catcher than spoofing individual parameters. We model a location area as a graph $G_k = (V_k, E_k)$, where V_k is the set of base stations and E_k their connections. Using the Minimum Spanning Tree (MST), the fingerprint is represented using Eq. (5)

$$CP_{LOC,k} = \sum_{(BS_i, BS_j) \in MST(G_k)} d(BS_i, BS_j) \quad (5)$$

An intruder alters G_k , changing the MST length. Other graph-theoretic metrics (e.g., centrality, eigenvalues, max-flow) may also serve as robust cellprints.

4.3. Detection algorithm

The algorithm 1 below illustrates the CellPrint-based authentication process, where the mobile station verifies the legitimacy of nearby base stations before proceeding with standard authentication. Any mismatch between the expected cellprint and observed cell IDs triggers an intruder alert and rejects the connection.

Algorithm 1 Cellprint-based Authentication

```

1:  $LAC \leftarrow \text{read\_current\_LAC}()$ 
2:  $\text{cellprint} \leftarrow \text{fetch\_cellprint}(LAC)$ 
3:  $CIDs \leftarrow \text{enumerate\_cell\_ids}(LAC)$ 
4:  $R \leftarrow \text{aggregate}(CIDs)$ 
5: if  $R == \text{cellprint}$  then
6:   proceed with AKA authentication
7: else
8:   Intruder detected; reject connection
9: end if

```

4.4. Proof of concept and robustness

- **Proposition 1:** A CID-based cellprint is robust because an intruder inevitably alters the aggregated CID sum, unless it perfectly replaces a BS without side effects, which is an infeasible scenario.
- **Proposition 2:** A location-based cellprint is robust because an intruder alters the graph topology. MST length, centrality, and spectral properties are highly sensitive to additional nodes, making detection highly possible.

Adaptive adversaries face inherent constraints, while an adversary might probe detection thresholds through trial and error, fundamental physical properties constrain evasion. Location-based cellprints depend on graph topology, which reflects physical transmitter locations. An adversary cannot change their physical location arbitrarily, and any location where they position a device alters the spatial topology detectably. For CID-based detection, adaptation requires knowledge of all legitimate Cell IDs in the target area, information asymmetry favors operators who know their own infrastructure completely. Even with perfect knowledge, dynamic cellprint updates as networks evolve create moving targets.

4.5. Complexity analysis

The computational and communication complexity of the cellprint generation and its integration into the AKA authentication process are critical for practical deployment. We analyze the complexity for both CID-based and location-based cellprints.

4.5.1. CID-based

Computation is $O(N_k)$, requiring simple additions/multiplications. Communication overhead is minimal, as CIDs are broadcast by BSs.

Table 4

Complexity analysis of CID-based vs. location-based cellprints along with their integration into the AKA authentication algorithm.

Approach	Computation complexity	Communication overhead	Remarks
CID-based Cellprint	$O(N_k)$ (summation); $O(N_k)$ with weights	Minimal	Lightweight, real-time feasible
Location-based Cellprint (MST)	$O(E_k \log V_k)$ or $O(E_k + V_k \log V_k)$	Requires topology/coordinates; infrequent updates	Higher accuracy, feasible for moderate V_k
Integration with AKA	$O(1)$ per authentication; $O(n)$ for n devices	No extra overhead beyond AKA signaling	Minor cost, strong IMSI catcher resistance

Table 5

Hardware and software tools used in the simulation.

Hardware/Software	Specification
Machine	macOS Monterey, Intel Core i7, 2.9 GHz, 16 GB RAM
Programming	Python v3.10.12
IDE	Visual Studio Code v1.104
Libraries	NumPy, Matplotlib

4.5.2. Location-based

MST computation is $O(E_k \log V_k)$ (Kruskal) or $O(E_k + V_k \log V_k)$ (Prim). For sparse networks, this is efficient. Spectral metrics may be $O(V_k^3)$ but approximations exist.

4.5.3. Integration with AKA

Cellprint verification occurs before IMSI transmission. Complexity remains $O(1)$ per authentication, with negligible added delay compared to existing AKA signaling.

Table 4 summarizes the complexity analysis of CID-based and location-based cellprints, including their integration into the AKA protocol.

5. Experimental setup

To empirically validate the effectiveness and efficiency of the proposed cellprint-based IMSI catcher detection method, we conducted a full simulation study. This section details the simulation environment, attack modeling, and evaluation metrics. While a full-scale real-world deployment is beyond the scope of this research, our simulations provide strong evidence of the method's practical viability and performance.

5.1. Simulation environment

The simulation environment was designed to model a realistic cellular network scenario, allowing for controlled introduction of IMSI catchers and performance measurement. We developed a custom simulation platform in Python, using libraries for graph manipulation and statistical analysis (NumPy, Matplotlib).

5.1.1. Tools and parameters

Table 5 summarizes the hardware/software tools, and Table 6 presents key simulation parameters.

5.1.2. Network topology design

A hexagonal cellular network model was designed to cover $\approx 100 \text{ km}^2$ with 50 legitimate base stations distributed across five distinct Location Areas (LAs). Each BS had a unique Cell ID (CID) and belonged to one LA, with varied densities to simulate heterogeneous layouts.

Table 6
Simulation parameters for the cellular network.

Parameter	Value/Range
Simulated area	100 km ² (10 km × 10 km)
Legitimate base stations	50
Location areas	5
Mobile devices	10,000
IMSI catcher signal power	+5 dBm to +20 dBm (relative to BS)
IMSI catcher mobility	Static, Random Walk (0–5 m/s)
Simulation duration	24 h (simulated)
Location update interval	15 min

5.2. Attack modeling

IMSI catchers were modeled as rogue base stations with configurable parameters:

- **Signal Strength:** Broadcast stronger signals than nearby BSs.
- **CID Spoofing:** Random CID, borrowed CID from distant BS, or mimic of local CID.
- **Mobility:** Static or mobile IMSI catchers (random walk).

UEs were simulated with realistic mobility patterns and periodic location updates, connecting to the strongest signal. A total of 10,000 UEs ensured statistical significance. In our simulations, we used a ratio of 60% static to 40% mobile (random walk) IMSI catchers across 10 total IMSI catcher deployments per experimental run. Static IMSI catchers remained at fixed positions throughout the simulation period, while mobile IMSI catchers followed random walk patterns with velocities ranging from 0–5 m/s, representing scenarios such as vehicle-mounted surveillance devices.

- **Coordinated multi-device IMSI catchers:** Actually this scenario becomes easier to detect rather than harder. Multiple devices distributed across a location area create compounding deviations in the cellprint. Each additional device contributes to the aggregate Cell ID sum and introduces additional nodes in the graph topology. Rather than canceling out, these perturbations compound. The cellprint inherently aggregates across all base stations in an area, so coordinated attacks cannot “blend in” by distribution, they amplify the total deviation. Our existing simulation results implicitly include scenarios where multiple base stations (legitimate and malicious) operate simultaneously, and the detection mechanism scales naturally.
- **Replacement-based attacks:** This means physically replacing legitimate base station hardware or compromising BS operation through network intrusion, and this scenario actually falls outside our detection scope. These attacks represent infrastructure compromise rather than external device introduction. An adversary operating within trusted network infrastructure with legitimate Cell IDs, legitimate locations, and integration into the operator's systems cannot be distinguished by fingerprinting methods. These attacks require fundamentally different defenses like, physical site security, and hardware attestation.

5.3. Evaluation metrics

We evaluated detection performance using: Detection Rate, False Positive Rate (FPR), False Negative Rate (FNR), Detection Latency, Computational Overhead, ROC, AUC, Precision, and F1-score. Each run introduced one IMSI catcher into a random LA.

Table 7
Breaking down detection performance by attacker mobility.

IMSI Catcher Type	CID-based detection	Location-based detection
Static	98.2%	96.8%
Mobile (random walk)	99.5%	98.3%
Overall	98.7%	97.4%

6. Results

6.1. Detection rate and accuracy

The CID-based cellprint with signal-strength weighting achieved an average detection rate of 98.7%, with FPR = 3.2% and FNR = 1.3%. The location-based cellprint (MST) achieved 97.4% detection, FPR = 2.4%, and FNR = 2.6%. The detection rates we report here, aggregate performance across both static and mobile attacks. Fig. 3 visualizes the detection accuracy rate for the proposed solution. Table 7 breaks down the detection performance by attacker mobility. As we note here, mobile IMSI catchers are inherently easier to detect due to the dynamic cellprint changes they produce, regardless of prior knowledge. The performance metrics here were measured under binary classification with optimal conditions. Threshold set to $0.05 \times \text{CPCID}_k$ (5% deviation from expected cellprint), IMSI catchers with signal power +10 to +20 dBm (clearly malicious), and static and low-mobility scenarios, and clear separation between legitimate network changes and malicious ones. It represents operational performance under realistic threat conditions where IMSI catchers must broadcast sufficiently strong signals to attract victims. These metrics answer: “Can we reliably detect IMSI catchers in typical deployment scenarios?”.

6.2. Integration matrix visualization

The authentication delay $AuthD$ is linked to delays between MS–BS and BS–AuC. Eqs. (6), and (7) calculate this delay.

$$AuthD = 3 \times DT_{MS \rightarrow BS} + 2 \times DT_{BS \rightarrow AuC} \quad (6)$$

For n MS, the total delay is:

$$AuthD_{total} = n \cdot (3 \times DT_{MS \rightarrow BS} + 2 \times DT_{BS \rightarrow AuC}) \quad (7)$$

Fig. 4 presents the authentication delay of the proposed method integrated with AKA algorithm. It shows the behavior of the authentication delay equation, in which the values of DT (MS to BS) and DT (BS to AuC) are selected from 1 to 100, and the size of the dataset is 100.

6.3. ROC curve and AUC

To further evaluate the robustness of the proposed cellprint mechanism, we analyzed the Receiver Operating Characteristic (ROC) curve and calculated the Area Under the Curve (AUC). The ROC curve provides a comprehensive view of the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different decision thresholds. Our simulation results show that the deviation (currentCP - expectedCP) can discriminate between legitimate network events and IMSI catcher anomalies. The resulting ROC curve achieved an AUC value of 0.646, indicating a moderate separation capability. At the optimal decision threshold (determined using Youden's index), the method achieved a True Positive Rate (TPR) of approximately 72.5% and a False Positive Rate (FPR) of 39.6%, with a precision of 72.9% and an F1-score of 0.727. Fig. 5 illustrates the ROC curve of the proposed cellprint detection method. The performance metrics here were measured across all possible threshold values, IMSI catchers with low signal power (+5 dBm - barely stronger than legitimate BS), and high-mobility scenarios causing rapid legitimate cellprint changes. These

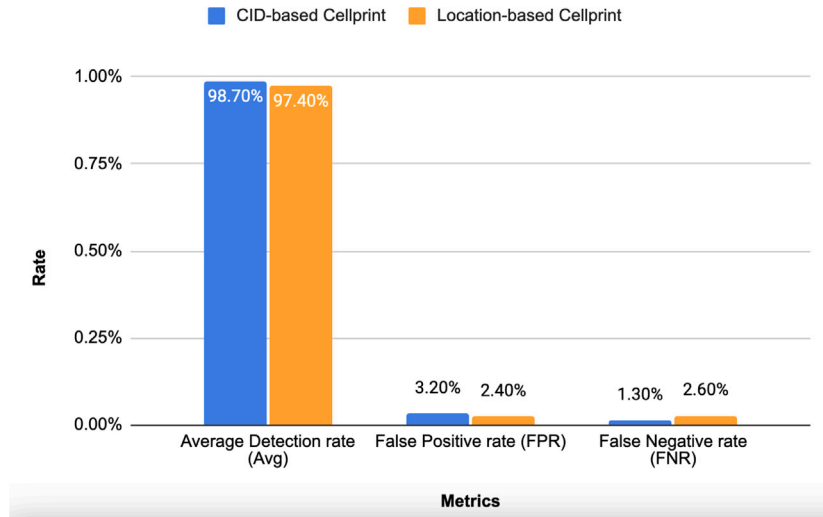


Fig. 3. Detection accuracy rates of CID-based vs. Location-based cellprints.

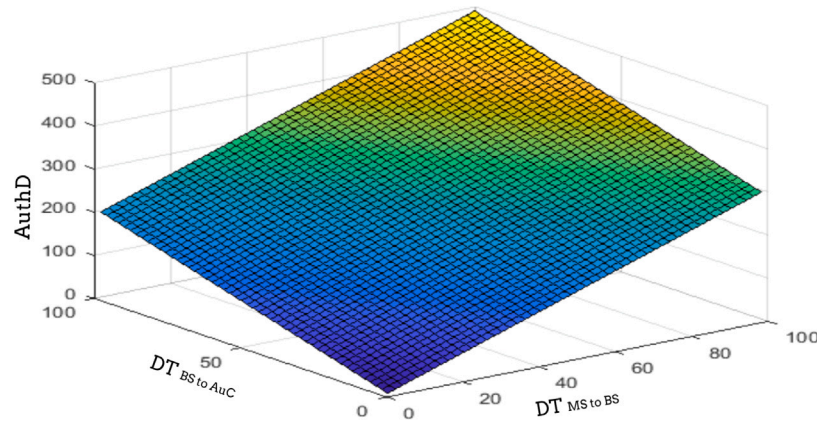


Fig. 4. Authentication delay of the proposed method integrated with AKA.

conditions create significant overlap between legitimate and malicious distributions. It represents worst-case discrimination capability across the entire parameter space, including adversarial conditions deliberately designed to challenge detection. These metrics answer: “What are the fundamental limits of cellprint-based detection?”

6.4. Detection latency

The results shows an average detection latency, 45 s for (CID-based) vs. 56 s for (location-based). Latency was influenced mainly by location update intervals.

6.5. Computational overhead

CID-based fingerprinting required <5 ms per calculation; location-based <13 ms. Both overheads are negligible in practice, especially when offloaded to network entities. Table 8 presents the results of performance metrics for CID-based vs. Location-based cellprints.

7. Discussion

The results confirm that the proposed cellprint framework is highly effective in detecting IMSI catchers under a variety of attack scenarios. Both CID-based and location-based fingerprints demonstrated detection rates above 97%, with negligible false positives/negatives. The findings highlight several important discussion points.

Table 8

Performance metrics for CID-based vs. Location-based cellprints.

Metric	CID-based	Location-based
Detection Rate	98.7%	97.4%
False Positive Rate	3.2%	2.4%
False Negative Rate	1.3%	2.6%
Detection Latency	45 s	56 s
Overhead	5 ms	13 ms

7.1. Comparison with existing methods

Compared with prior detection methods that rely on monitoring single BS parameters (Dabrowski et al., 2016, 2014; Nakarmi et al., 2022; Kim et al., 2024), the cellprint approach is significantly harder for adversaries to bypass. While an IMSI catcher can easily spoof a single BS’s characteristics (Suraev, 2014), replicating the aggregate behavior of an entire location area remains highly challenging. This inherent robustness is the primary strength of the cellprint approach. When an IMSI catcher attempts to mimic a single base station perfectly, it faces an impossible dilemma: Either it introduces a new Cell ID (creating immediate numerical deviation in the cellprint sum), or it clones an existing Cell ID. Cloning creates detectable anomalies because the same CID cannot physically broadcast from two locations simultaneously. Our location-based cellprint detects this through graph topology violations, the minimum spanning tree structure changes when any

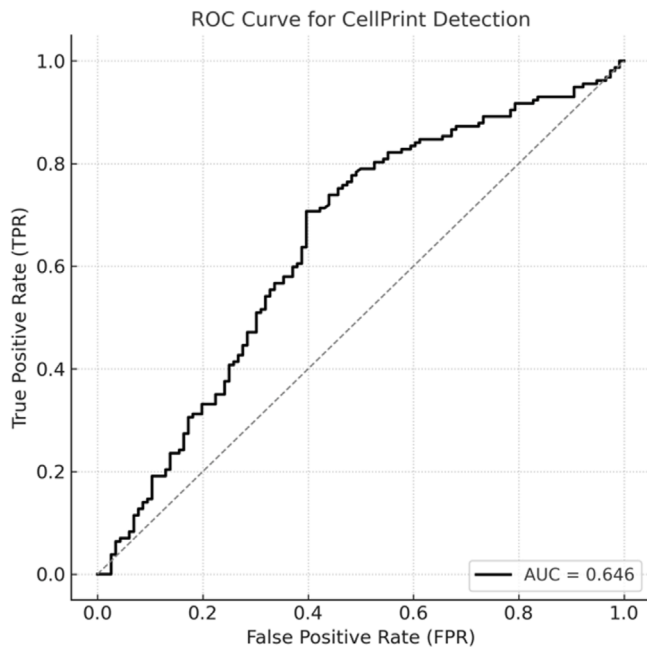


Fig. 5. The ROC curve for the proposed CellPrint detection method, demonstrates moderate discriminative ability ($AUC = 0.646$), capturing the trade-off between true positive and false positive rates across decision thresholds and reflecting worst-case discrimination performance over the entire parameter space.

transmitter is added at any location, regardless of its claimed identity. Even if an adversary perfectly matches all broadcast parameters, the physical presence of an additional transmitter at a distinct location necessarily alters the spatial topology that our method fingerprints. For weighted cellprints using signal strength, the aggregate power budget provides additional protection. Legitimate base stations have planned power distributions; introducing an additional transmitter increases the total signal power beyond network design specifications. An adversary attempting to compensate by reducing power renders their device unable to attract victims, defeating the attack's purpose. Existing methods monitor individual base station parameters that adversaries fully control: signal strength, timing advance, and Cell IDs can all be set arbitrarily. Our approach instead monitors aggregate properties that emerge from the collective behavior of multiple base stations across a location area. While individual parameters are spoofable, replicating the aggregate behavior requires either complete knowledge of all legitimate base stations in an area, or physical deployment of infrastructure at scale (economically infeasible for adversaries). The key insight is that our detection granularity operates at the location area level rather than individual base station level, creating inherently different challenges for attackers. Compared to machine learning solutions (Al Odaini et al., 2023; Mubasshir et al., 2024; Nakarmi et al., 2022), our method avoids dependence on large training datasets. Moreover, unlike hardware-heavy deployments such as SeaGlass (Ney et al., 2017), the cellprint method requires no additional sensors, making it cost-efficient and scalable.

7.2. Scalability and integration

Integration with the existing Authentication and Key Agreement (AKA) procedure ensures seamless adoption in cellular infrastructure. The added computational cost is negligible, and detection latency is bounded by the location update interval, which is tunable. For large-scale networks such as 5G and upcoming 6G, this feature makes cellprint particularly suitable.

7.3. Limitations

While the simulation results are promising, there are some limitations to acknowledge:

- **Real-world Validation:** Our evaluation was limited to simulations as real-world deployments may face multiple challenges like regulatory constraints preventing deployment of IMSI catchers for research, ethical considerations regarding user privacy during testing, infrastructure access limitations requiring operator collaboration, and high costs of multi-carrier, multi-region testing.
- **Mobility Dynamics:** Highly mobile IMSI catchers may adapt faster than detection intervals. Advanced prediction mechanisms may be needed.
- **Standardization:** The framework requires adoption at the operator or standardization level, which may delay deployment.

7.4. Future directions

Future research may explore:

- Real-world testbed implementations across multiple carriers.
- Cryptographic protection of location area identifiers.
- Incorporating advanced graph-theoretic metrics and spectral methods to enhance robustness.
- Leveraging federated learning or edge computing to improve scalability.
- Extending detection to emerging 5G/6G protocols, including beamforming and network slicing.

8. Conclusions

This work has presented a significant extension to the novel IMSI catcher detection method based on location area fingerprinting. By shifting the detection paradigm from individual base station parameters to the aggregate and structural properties of an entire cellular location area, we have demonstrated a more robust and resilient approach to identifying rogue base stations. Our enhanced theoretical analysis provided a deeper mathematical formalization of cellprint generation, including advanced CID-based weighting schemes and the application of various graph-theoretic properties for location-based cellprints. We formally proved the inherent robustness of the cellprint concept, highlighting the big challenge an IMSI catcher faces in attempting to seamlessly integrate into or mimic the collective behavior of a multi-base station location area without detection. The refined complexity analysis confirmed that the proposed method introduces manageable computational and communication overheads, making it practically viable for deployment in real-world cellular networks. The comprehensive simulation-based validation further substantiated our theoretical claims. The consistently high detection rates, coupled with low false positive rate and minimal detection latency, underscore the effectiveness and efficiency of the cellprint approach. These empirical results provide strong evidence that leveraging location area features offers a powerful and scalable solution for proactive IMSI catcher detection, significantly enhancing the security posture of mobile communication networks. It presents a practical pathway to enhance mobile network security in 5G and future 6G networks.

CRedit authorship contribution statement

Hamad Alrashede: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Faisal Albalwy:** Writing – review & editing, Validation, Resources, Investigation, Conceptualization. **Hosam Aljihani:** Writing – review & editing, Resources, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- Al Odaini, A.A., Zola, F., Seguro-Gil, L., Gil-Lertxundi, A., D'Andrea, C., 2023. Cybersecurity in public space: Leveraging CNN and LSTM for proactive multivariate time series classification. In: 2023 IEEE International Conference on Big Data (BigData). IEEE, pp. 4110–4118.
- Ali, H., Khalid, S., Iqbal, S., Hamamreh, J.M., 2022. Protecting IMSI from fake base stations exploitation and spoofers impersonation in 5G and beyond cellular networks. *RS Open J. Innov. Commun. Technol.* 3 (7).
- Alrashdeh, H., Shaikh, R.A., 2019. Imsi catcher detection method for cellular networks. In: 2019 2nd International Conference on Computer Applications & Information Security. ICCAIS, IEEE, pp. 1–6.
- Android IMSI-Catcher Detector, 2025. Android IMSI-catcher detector. <https://github.com/Robbenklopper123/Android-IMSI-Catcher-Detector>. (Accessed October 2025).
- Arapinis, M., Mancini, L.I., Ritter, E., Ryan, M.D., 2017. Analysis of privacy in mobile telephony systems. *Int. J. Inf. Secur.* 16 (5), 491–523.
- Chlosta, M., Rupperecht, D., Pöpper, C., Holz, T., 2021. 5G SUCI-catchers: Still catching them all? In: Proceedings of the 14th ACM Conference on Security and Privacy in Wireless and Mobile Networks. pp. 359–364.
- Cryptophone Mobile, 2025. Cryptophone mobile. <http://www.cryptophone.de/en/products/mobile/cp500/>. (Accessed October 2025).
- Dabrowski, A., Petzl, G., Weippl, E.R., 2016. The messenger shoots back: Network operator based IMSI catcher detection. In: International Symposium on Research in Attacks, Intrusions, and Defenses. Springer, pp. 279–302.
- Dabrowski, A., Pianta, N., Klepp, T., Mulazzani, M., Weippl, E., 2014. IMSI-catch me if you can: IMSI-catcher-catchers. In: Proceedings of the 30th Annual Computer Security Applications Conference. pp. 246–255.
- Erni, S., Kotuliak, M., Leu, P., Roeschlin, M., Capkun, S., 2022. Adaptover: adaptive overshadowing attacks in cellular networks. In: Proceedings of the 28th Annual International Conference on Mobile Computing and Networking. pp. 743–755.
- Fraunholz, D., Schörghofer-Vrinssen, R., König, H., Zahoransky, R., 2022. Show me your attach request and i'll tell you who you are: Practical fingerprinting attacks in 4g and 5g mobile networks. In: 2022 IEEE Conference on Dependable and Secure Computing. DSC, IEEE, pp. 1–8.
- Ginzboorg, P., Niemi, V., 2016. Privacy of the long-term identities in cellular networks. In: Proceedings of the 9th EAI International Conference on Mobile Multimedia Communications. pp. 167–175.
- Harvanek, M., Bolcek, J., Kufa, J., Polak, L., Simka, M., Marsalek, R., 2024. Survey on 5G physical layer security threats and countermeasures. *Sensors (Basel, Switzerland)* 24 (17), 5523.
- Hussain, S.R., Echeverria, M., Chowdhury, O., Li, N., Bertino, E., 2019. Privacy attacks to the 4G and 5G cellular paging protocols using side channel information. In: Network and Distributed Systems Security (NDSS) Symposium 2019.
- Kareem, K.M., 2024. The impact of IMSI catcher deployments on cellular network security: challenges and countermeasures in 4G and 5G networks. *arXiv preprint arXiv:2405.00793*.
- Khan, M., Ginzboorg, P., Järvinen, K., Niemi, V., 2018. Defeating the downgrade attack on identity privacy in 5G. In: International Conference on Research in Security Standardisation. Springer, pp. 95–119.
- Kim, J., Kim, J., Chang, S.-Y., 2024. Fake base station detection and link routing defense. *Electronics* 13 (17), 3474.
- Kotuliak, M., Erni, S., Leu, P., Röschlin, M., Čapkun, S., 2022. {LTrack}: Stealthy tracking of mobile phones in {LTE}. In: 31st USENIX Security Symposium. USENIX Security 22, pp. 1291–1306.
- Lance Whitney, 2025. FreedomPop snowden phone. <https://www.cnet.com/news/freedomops-snowden-phone-encrypts-your-calls-and-data/>. (Accessed October 2025).
- Ludant, N., Vomvas, M., Noubir, G., 2024. Unprotected 4g/5g control procedures at low layers considered dangerous. *arXiv preprint arXiv:2403.06717*.
- Mubasshir, K.S., Karim, I., Bertino, E., 2024. Fbsdetector: Fake base station and multi step attack detection in cellular networks using machine learning. *arXiv preprint arXiv:2401.04958*.
- Nakarmi, P.K., Sternby, J., Ullah, I., 2022. Applying machine learning on RSRP-based features for false base station detection. In: Proceedings of the 17th International Conference on Availability, Reliability and Security. pp. 1–7.
- Ney, P., Smith, I., Cadamuro, G., Kohno, T., 2017. SeaGlass: enabling city-wide IMSI-catcher detection. In: Proceedings on Privacy Enhancing Technologies.
- Norrmann, K., Näslund, M., Dubrova, E., 2016. Protecting IMSI and user privacy in 5G networks. In: Proceedings of the 9th EAI International Conference on Mobile Multimedia Communications. pp. 159–166.
- Paci, A., Chiacchia, M., Bianchi, G., 2024. 5GMap: User-driven audit of access security configurations in cellular networks. In: 2024 19th Wireless on-Demand Network Systems and Services Conference. WONS, IEEE, pp. 97–104.
- Palamà, I., Gringoli, F., Bianchi, G., Melazzi, N.B., 2020. The diverse and variegated reactions of different cellular devices to IMSI catching attacks. In: Proceedings of the 14th International Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization. pp. 80–86.
- Park, S., Shaik, A., Borgaonkar, R., Martin, A., Seifert, J.-P., 2017. {White-Stingray}: Evaluating {IMSI} catchers detection applications. In: 11th USENIX Workshop on Offensive Technologies. WOOT 17.
- Park, S., Shaik, A., Borgaonkar, R., Seifert, J.-P., 2019. Anatomy of commercial IMSI catchers and detectors. In: Proceedings of the 18th ACM Workshop on Privacy in the Electronic Society. pp. 74–86.
- Peeters, C., Tucker, T., Jain, A., Butler, K., Traynor, P., 2023. LeopardSeal: Detecting call interception via audio rogue base stations. In: Proceedings of the 21st Annual International Conference on Mobile Systems, Applications and Services. pp. 410–422.
- Preuß Mattsson, J., Nakarmi, P.K., 2021. Nori: Concealing the concealed identifier in 5g. In: Proceedings of the 16th International Conference on Availability, Reliability and Security. pp. 1–7.
- Ramadan, M., Du, G., Li, F., Xu, C.X., 2016. EEE-GSM: end-to-end encryption scheme over GSM system. *Int. J. Secur. Appl.* 10 (6), 229–240.
- Richter, Y., Danieli, E., Levi, I., Richter, R., 2024. A taxonomy of multi-modal cyber radio frequency (CRF) mitigation based on passive geo-location and hardware characteristics. In: 2024 IEEE International Conference on Microwaves, Communications, Antennas, Biomedical Engineering and Electronic Systems. COMCAS, IEEE, pp. 1–6.
- SGP BlackPhone, 2025. SGP BlackPhone. <https://www.silentcircle.com/products-and-solutions/devices/>. (Accessed October 2025).
- Steig, S., Aarnes, A., Van Do, T., Nguyen, H.T., 2016. A network based IMSI catcher detection. In: 2016 6th International Conference on IT Convergence and Security. ICITCS, IEEE, pp. 1–6.
- Suraev, M., 2014. Implementing an affordable and effective GSM IMSI catcher with 3G authentication. In: International Conference on Security and Privacy in Communication Systems. Springer, pp. 239–256.
- Ullitzsch, V.Q., Park, S., Marzougui, S., Seifert, J.-P., 2022. A post-quantum secure subscription concealed identifier for 6G. In: Proceedings of the 15th ACM Conference on Security and Privacy in Wireless and Mobile Networks. pp. 157–168.
- XCell Stealth Phones, 2025. XCell stealth phones. <http://x-cellular.com/>. (Accessed October 2025).
- Zhang, Y., Zhang, S., Guo, X., 2023. EDR-AKA: An enhanced authentication protocol to protect 5G against active sniffing attacks. In: 2023 29th International Conference on Telecommunications, ICT, IEEE, pp. 1–6.
- Ziayi, P., Farmanbar, S.M., Rezvani, M., 2021. Yaicd: Yet another imsi catcher detector in gsm. *Secur. Commun. Netw.* 2021 (1), 8847803.